08_Midterm_Project_Yulia_Chernysheva

October 16, 2026

Yuliia Chernysheva
October 16, 2026
CIS 2267 Midterm Project: US Farmland Data Analysis (2012 vs 2017)

[1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Set standard plotting aesthetics for Seaborn
sns.set_theme(style="whitegrid")
```

1 PART 1: Loading, Cleaning, and Analyzing Farmland Census Data

[2]:
```
# 1. Load the initial dataset from CSV into a DataFrame
df_farmland = pd.read_csv("Farmland.csv")

# Display initial rows and dataset summary
print("=> Initial five rows of the Farmland dataset:\n")
display(df_farmland.head())

print("\n=> Dataset Structure and Info:\n")
display(df_farmland.info())
```

```
=> Initial five rows of the Farmland dataset:
```

	State	2012Farmland	2017Farmland
0	Alabama	8,902,654	8,580,940
1	Alaska	833,861	849,753
2	Arizona	26,249,195	26,125,819
3	Arkansas	13,810,786	13,888,929
4	California	25,569,001	24,522,801

```
=> Dataset Structure and Info:

<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 50 entries, 0 to 49
Data columns (total 3 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   State          50 non-null     object
 1   2012Farmland   50 non-null     object
 2   2017Farmland   50 non-null     object
dtypes: object(3)
memory usage: 1.3+ KB
```

```
None
```

[3]:
```
# 2. Rename columns to standard snake_case for consistency
df_farmland = df_farmland.rename(columns={
    'State':'state',
    '2012Farmland':'farmland_2012',
    '2017Farmland':'farmland_2017'
})

# Normalize state names to Title Case (handles 'OHIO' -> 'Ohio')
df_farmland.state = df_farmland.state.str.title()

# Clean string formatting (remove commas/spaces) and convert to integer values
for col in ['farmland_2012', 'farmland_2017']:
    if df_farmland[col].dtype == 'object':
        df_farmland[col] = (
            df_farmland[col]
                .astype(str)
                .str.replace(',','', regex=False)
                .str.strip()
                .astype(int)
        )

# 3. Calculate Absolute Difference (Net Change in Acres) and Relative␣
 ↪Difference (% Change)
df_farmland['diff_abs'] = (df_farmland.farmland_2017 - df_farmland.
 ↪farmland_2012)
df_farmland['diff_pct'] = ((df_farmland.diff_abs/df_farmland.farmland_2012) *␣
 ↪100)

# 4. Sort Rankings
print("=> Top States by Absolute Farmland Change (Highest Gain to Largest Loss):
 ↪\n")
df_farmland.sort_values(
    by='diff_abs',
    ascending=False
).assign(
```

```
    farmland_2012=lambda df: df['farmland_2012'].map('{:,.0f}'.format),
    farmland_2017=lambda df: df['farmland_2017'].map('{:,.0f}'.format),
    diff_abs=lambda df: df['diff_abs'].map('{:+,.0f}'.format),
    diff_pct=lambda df: df['diff_pct'].map('{:+.2f}%'.format)
)
```

=> Top States by Absolute Farmland Change (Highest Gain to Largest Loss):

[3]:

	state	farmland_2012	farmland_2017	diff_abs	diff_pct
9	Georgia	9,620,836	9,953,730	+332,894	+3.46%
13	Indiana	14,720,396	14,969,996	+249,600	+1.70%
27	Nevada	5,913,761	6,128,153	+214,392	+3.63%
8	Florida	9,548,342	9,731,731	+183,389	+1.92%
17	Louisiana	7,900,864	7,997,511	+96,647	+1.22%
33	North Dakota	39,262,613	39,341,591	+78,978	+0.20%
3	Arkansas	13,810,786	13,888,929	+78,143	+0.57%
12	Illinois	26,937,721	27,006,288	+68,567	+0.25%
47	West Virginia	3,606,674	3,662,178	+55,504	+1.54%
29	New Jersey	715,057	734,084	+19,027	+2.66%
7	Delaware	508,652	525,324	+16,672	+3.28%
1	Alaska	833,861	849,753	+15,892	+1.91%
32	North Carolina	8,414,756	8,430,522	+15,766	+0.19%
41	Tennessee	10,867,812	10,874,238	+6,426	+0.06%
10	Hawaii	1,129,317	1,135,352	+6,035	+0.53%
34	Ohio	13,960,604	13,965,295	+4,691	+0.03%
38	Rhode Island	69,589	56,864	-12,725	-18.29%
40	South Dakota	43,257,079	43,243,742	-13,337	-0.03%
20	Massachusetts	523,517	491,653	-31,864	-6.09%
19	Maryland	2,030,745	1,990,122	-40,623	-2.00%
28	New Hampshire	474,065	425,393	-48,672	-10.27%
6	Connecticut	436,539	381,539	-55,000	-12.60%
44	Vermont	1,251,713	1,193,437	-58,276	-4.66%
14	Iowa	30,622,731	30,563,878	-58,853	-0.19%
5	Colorado	31,886,676	31,820,957	-65,719	-0.21%
11	Idaho	11,760,109	11,691,912	-68,197	-0.58%
46	Washington	14,748,107	14,679,857	-68,250	-0.46%
16	Kentucky	13,049,347	12,961,784	-87,563	-0.67%
2	Arizona	26,249,195	26,125,819	-123,376	-0.47%
18	Maine	1,454,104	1,307,613	-146,491	-10.07%
43	Utah	10,974,396	10,811,604	-162,792	-1.48%
21	Michigan	9,948,564	9,764,090	-184,474	-1.85%
35	Oklahoma	34,356,110	34,156,290	-199,820	-0.58%
39	South Carolina	4,971,244	4,744,913	-226,331	-4.55%
48	Wisconsin	14,568,926	14,318,630	-250,296	-1.72%
31	New York	7,183,576	6,866,171	-317,405	-4.42%
0	Alabama	8,902,654	8,580,940	-321,714	-3.61%

36	Oregon	16,301,578	15,962,322	-339,256	-2.08%
26	Nebraska	45,331,783	44,986,821	-344,962	-0.76%
15	Kansas	46,137,295	45,759,319	-377,976	-0.82%
37	Pennsylvania	7,704,444	7,278,668	-425,776	-5.53%
24	Missouri	28,266,137	27,781,883	-484,254	-1.71%
45	Virginia	8,302,444	7,797,979	-504,465	-6.08%
23	Mississippi	10,931,080	10,415,136	-515,944	-4.72%
22	Minnesota	26,035,838	25,516,982	-518,856	-1.99%
4	California	25,569,001	24,522,801	-1,046,200	-4.09%
49	Wyoming	30,363,641	29,004,884	-1,358,757	-4.47%
25	Montana	59,758,917	58,122,878	-1,636,039	-2.74%
30	New Mexico	43,201,023	40,659,836	-2,541,187	-5.88%
42	Texas	130,153,438	127,036,184	-3,117,254	-2.40%

[4]:
```
print("=> Top States by Percentage Farmland Change (Highest Gain to Largest␣
 ↪Loss):\n")
df_farmland.sort_values(
    by='diff_pct',
    ascending=False
).assign(
    farmland_2012=lambda df: df['farmland_2012'].map('{:,.0f}'.format),
    farmland_2017=lambda df: df['farmland_2017'].map('{:,.0f}'.format),
    diff_abs=lambda df: df['diff_abs'].map('{:+,.0f}'.format),
    diff_pct=lambda df: df['diff_pct'].map('{:+.2f}%'.format)
)
```

```
=> Top States by Percentage Farmland Change (Highest Gain to Largest Loss):
```

[4]:

	state	farmland_2012	farmland_2017	diff_abs	diff_pct
27	Nevada	5,913,761	6,128,153	+214,392	+3.63%
9	Georgia	9,620,836	9,953,730	+332,894	+3.46%
7	Delaware	508,652	525,324	+16,672	+3.28%
29	New Jersey	715,057	734,084	+19,027	+2.66%
8	Florida	9,548,342	9,731,731	+183,389	+1.92%
1	Alaska	833,861	849,753	+15,892	+1.91%
13	Indiana	14,720,396	14,969,996	+249,600	+1.70%
47	West Virginia	3,606,674	3,662,178	+55,504	+1.54%
17	Louisiana	7,900,864	7,997,511	+96,647	+1.22%
3	Arkansas	13,810,786	13,888,929	+78,143	+0.57%
10	Hawaii	1,129,317	1,135,352	+6,035	+0.53%
12	Illinois	26,937,721	27,006,288	+68,567	+0.25%
33	North Dakota	39,262,613	39,341,591	+78,978	+0.20%
32	North Carolina	8,414,756	8,430,522	+15,766	+0.19%
41	Tennessee	10,867,812	10,874,238	+6,426	+0.06%
34	Ohio	13,960,604	13,965,295	+4,691	+0.03%
40	South Dakota	43,257,079	43,243,742	-13,337	-0.03%

14	Iowa	30,622,731	30,563,878	-58,853	-0.19%
5	Colorado	31,886,676	31,820,957	-65,719	-0.21%
46	Washington	14,748,107	14,679,857	-68,250	-0.46%
2	Arizona	26,249,195	26,125,819	-123,376	-0.47%
11	Idaho	11,760,109	11,691,912	-68,197	-0.58%
35	Oklahoma	34,356,110	34,156,290	-199,820	-0.58%
16	Kentucky	13,049,347	12,961,784	-87,563	-0.67%
26	Nebraska	45,331,783	44,986,821	-344,962	-0.76%
15	Kansas	46,137,295	45,759,319	-377,976	-0.82%
43	Utah	10,974,396	10,811,604	-162,792	-1.48%
24	Missouri	28,266,137	27,781,883	-484,254	-1.71%
48	Wisconsin	14,568,926	14,318,630	-250,296	-1.72%
21	Michigan	9,948,564	9,764,090	-184,474	-1.85%
22	Minnesota	26,035,838	25,516,982	-518,856	-1.99%
19	Maryland	2,030,745	1,990,122	-40,623	-2.00%
36	Oregon	16,301,578	15,962,322	-339,256	-2.08%
42	Texas	130,153,438	127,036,184	-3,117,254	-2.40%
25	Montana	59,758,917	58,122,878	-1,636,039	-2.74%
0	Alabama	8,902,654	8,580,940	-321,714	-3.61%
4	California	25,569,001	24,522,801	-1,046,200	-4.09%
31	New York	7,183,576	6,866,171	-317,405	-4.42%
49	Wyoming	30,363,641	29,004,884	-1,358,757	-4.47%
39	South Carolina	4,971,244	4,744,913	-226,331	-4.55%
44	Vermont	1,251,713	1,193,437	-58,276	-4.66%
23	Mississippi	10,931,080	10,415,136	-515,944	-4.72%
37	Pennsylvania	7,704,444	7,278,668	-425,776	-5.53%
30	New Mexico	43,201,023	40,659,836	-2,541,187	-5.88%
45	Virginia	8,302,444	7,797,979	-504,465	-6.08%
20	Massachusetts	523,517	491,653	-31,864	-6.09%
18	Maine	1,454,104	1,307,613	-146,491	-10.07%
28	New Hampshire	474,065	425,393	-48,672	-10.27%
6	Connecticut	436,539	381,539	-55,000	-12.60%
38	Rhode Island	69,589	56,864	-12,725	-18.29%

[5]:
```
# 5. Descriptive Statistics for Numerical Columns
numeric_cols = ['farmland_2012','farmland_2017','diff_abs','diff_pct']
summary_stats = df_farmland[numeric_cols].agg(['sum', 'mean', 'min', 'max'])

print("\n=> Summary Statistics (Sum, Mean, Min, Max):")
display(summary_stats.assign(
    farmland_2012=lambda df: df['farmland_2012'].map('{:,.0f}'.format),
    farmland_2017=lambda df: df['farmland_2017'].map('{:,.0f}'.format),
    diff_abs=lambda df: df['diff_abs'].map('{:+,.0f}'.format),
    diff_pct=lambda df: df['diff_pct'].map('{:+.2f}%'.format)
))

# The true national percentage change
```

```
total_2012 = df_farmland['farmland_2012'].sum()
total_2017 = df_farmland['farmland_2017'].sum()
national_pct_change = ((total_2017 - total_2012) / total_2012) * 100

print(f"\n=> True Overall National Farmland Change (2012-2017):␣
 ↪{national_pct_change:.2f}%")
```

```
=> Summary Statistics (Sum, Mean, Min, Max):
```

	farmland_2012	farmland_2017	diff_abs	diff_pct
sum	914,527,657	900,217,576	-14,310,081	-104.93%
mean	18,290,553	18,004,352	-286,202	-2.10%
min	69,589	56,864	-3,117,254	-18.29%
max	130,153,438	127,036,184	+332,894	+3.63%

```
=> True Overall National Farmland Change (2012-2017): -1.56%
```

Note on Statistical Aggregation:

- `Sum of percentages (-104.93%):` Summing percentage changes across states is mathematically invalid because each percentage represents a change relative to a different state base.
- `State Mean (-2.10%):` The average percentage change across all 50 states treats every state equally regardless of physical size.
- `True National Change (-1.56%):` The actual national loss of agricultural land (-14.31 million acres) calculated directly from total 2012 and 2017 farmland totals.

[6]:
```
# 6. Visualization: Four Horizontal Bar Charts
titles_dict = {
    'farmland_2012': 'Total Farmland Area in 2012 (Acres)',
    'farmland_2017': 'Total Farmland Area in 2017 (Acres)',
    'diff_abs': 'Absolute Change in Farmland (2012 to 2017, Acres)',
    'diff_pct': 'Relative Percentage Change in Farmland (2012 to 2017, %)'
}

for col in numeric_cols:
    plt.figure(figsize=(12,14))

    # Sort dataset descending for each metric
    sorted_df  = df_farmland[[col,'state']].sort_values(by=col, ascending=False)

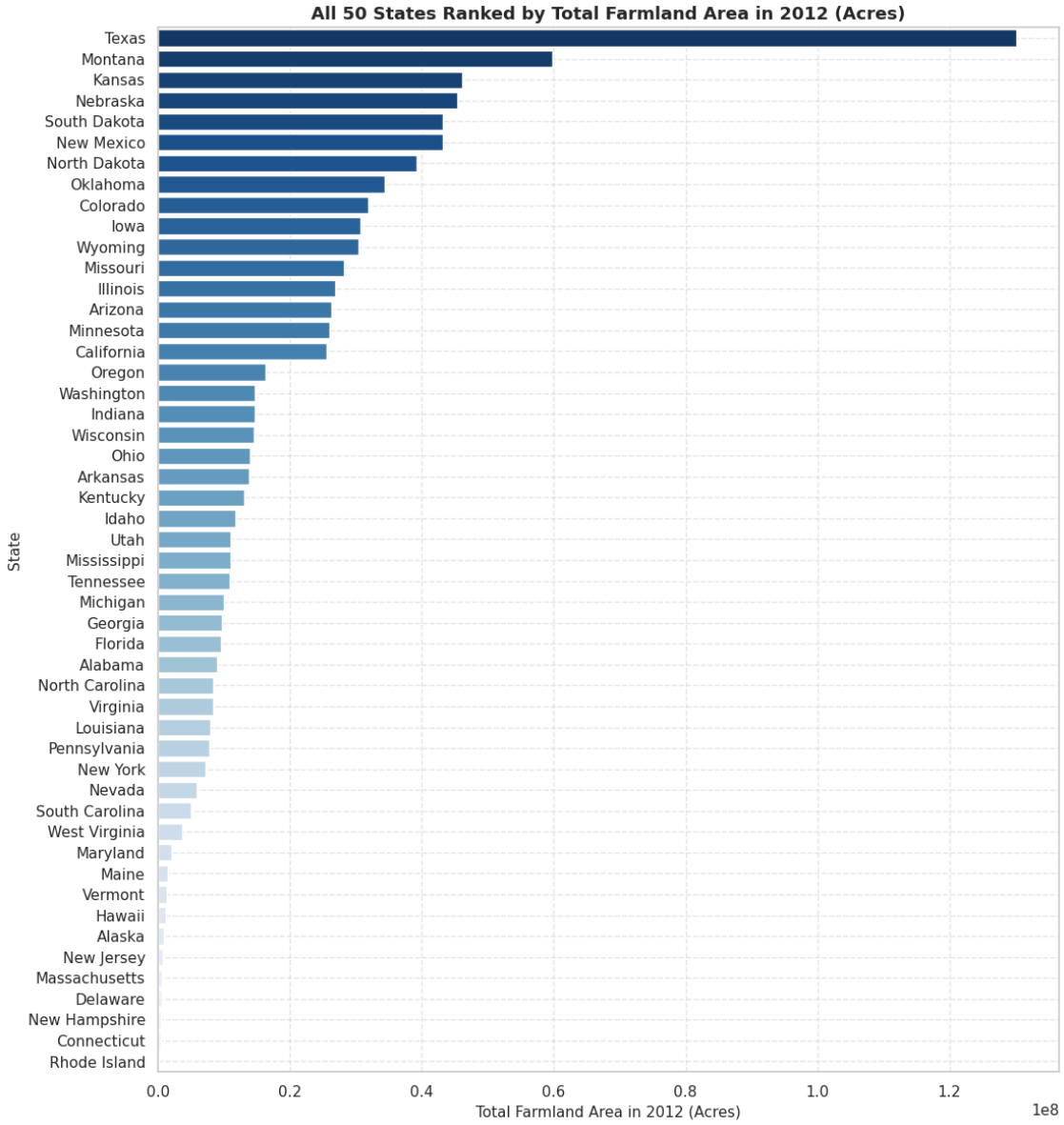
    # Plot horizontal bars
    sns.barplot(
        data=sorted_df,
        x=col,
        y='state',
        palette='Blues_r' if 'diff' not in col else 'coolwarm',
```

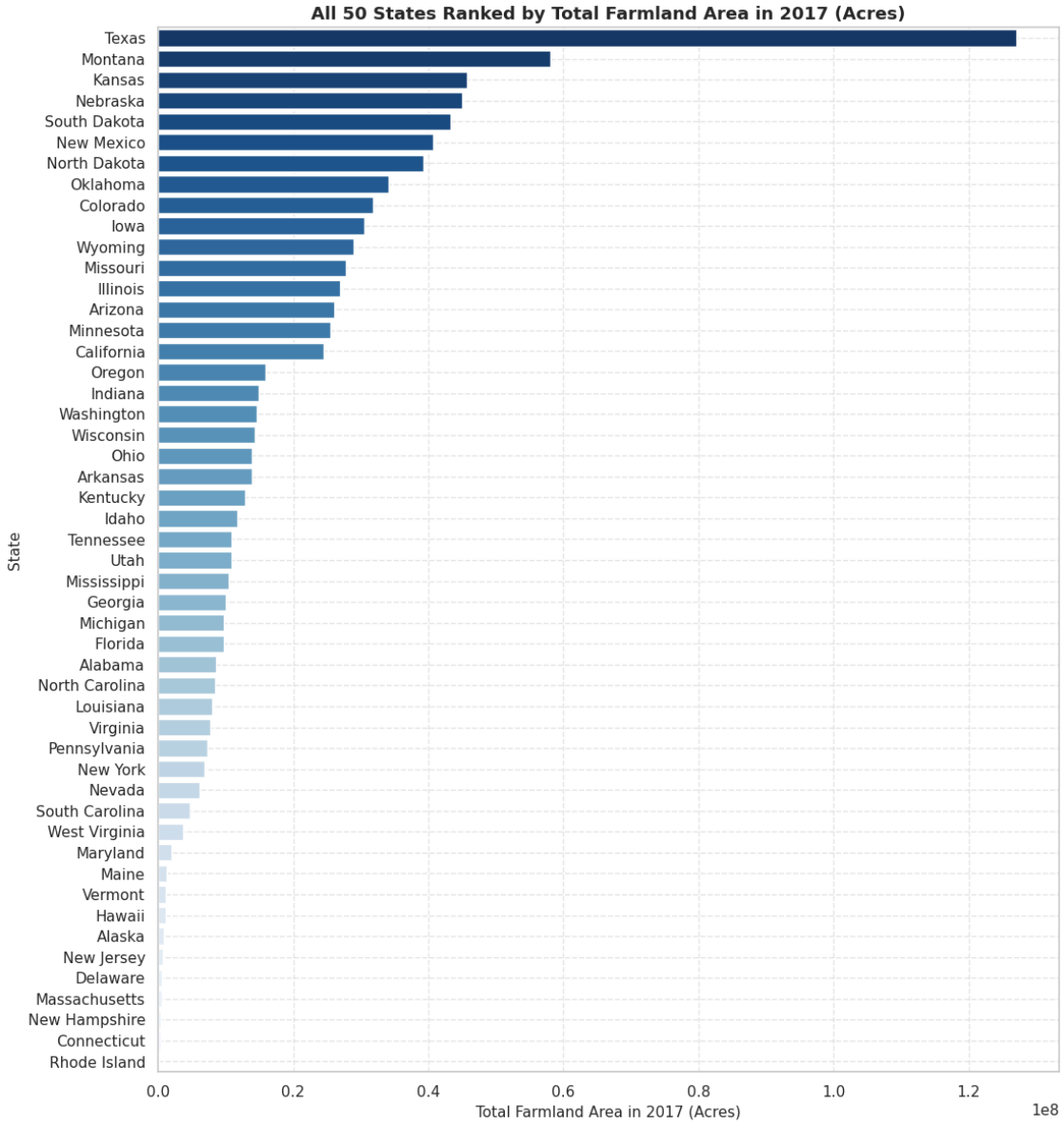
```
        hue='state',
        legend=False)

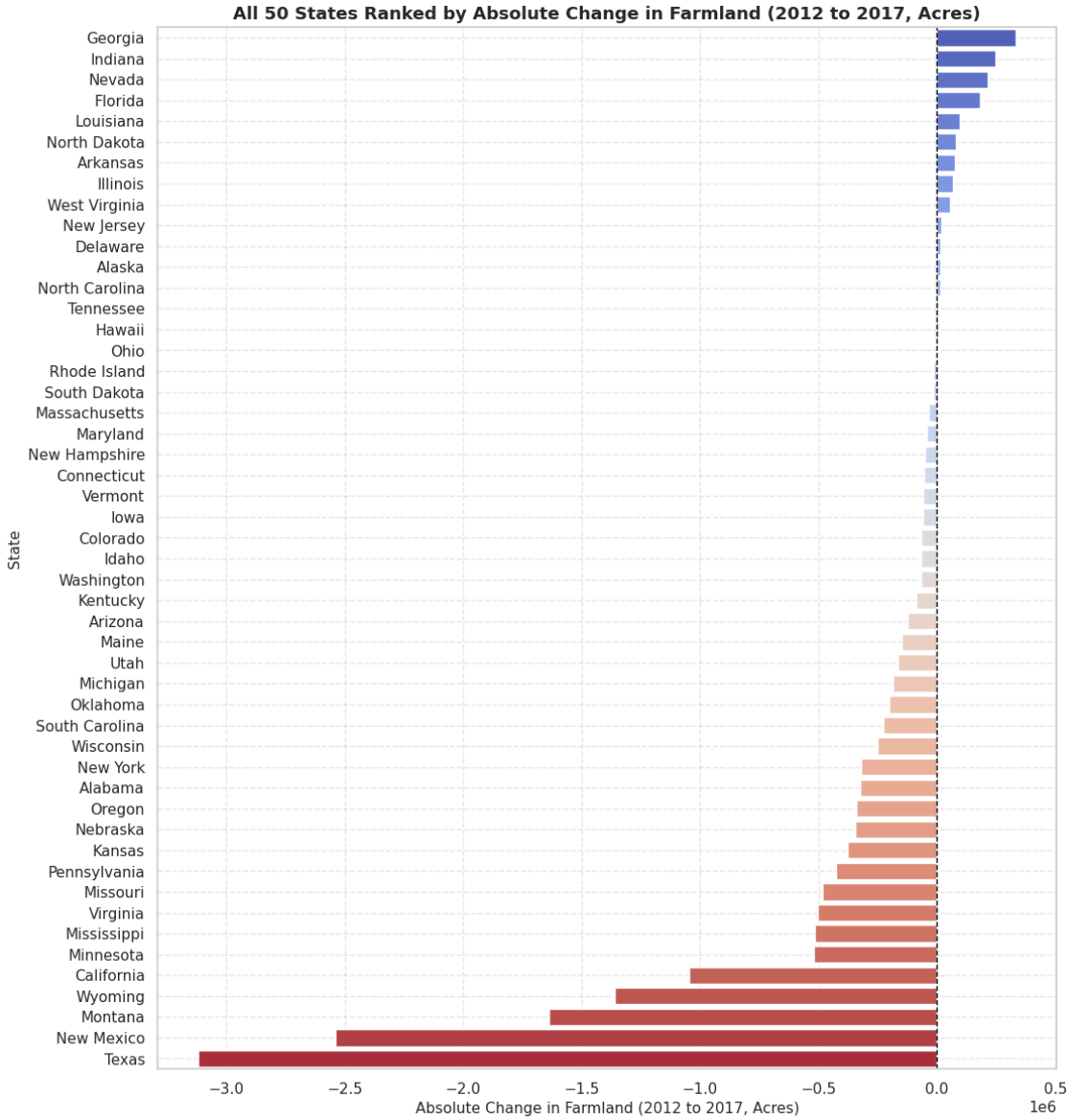
    # Add vertical zero line for change metrics to clearly separate gains and␣
 ↪losses
    if 'diff' in col:
        plt.axvline(0, color='black', linewidth=1, linestyle='--')

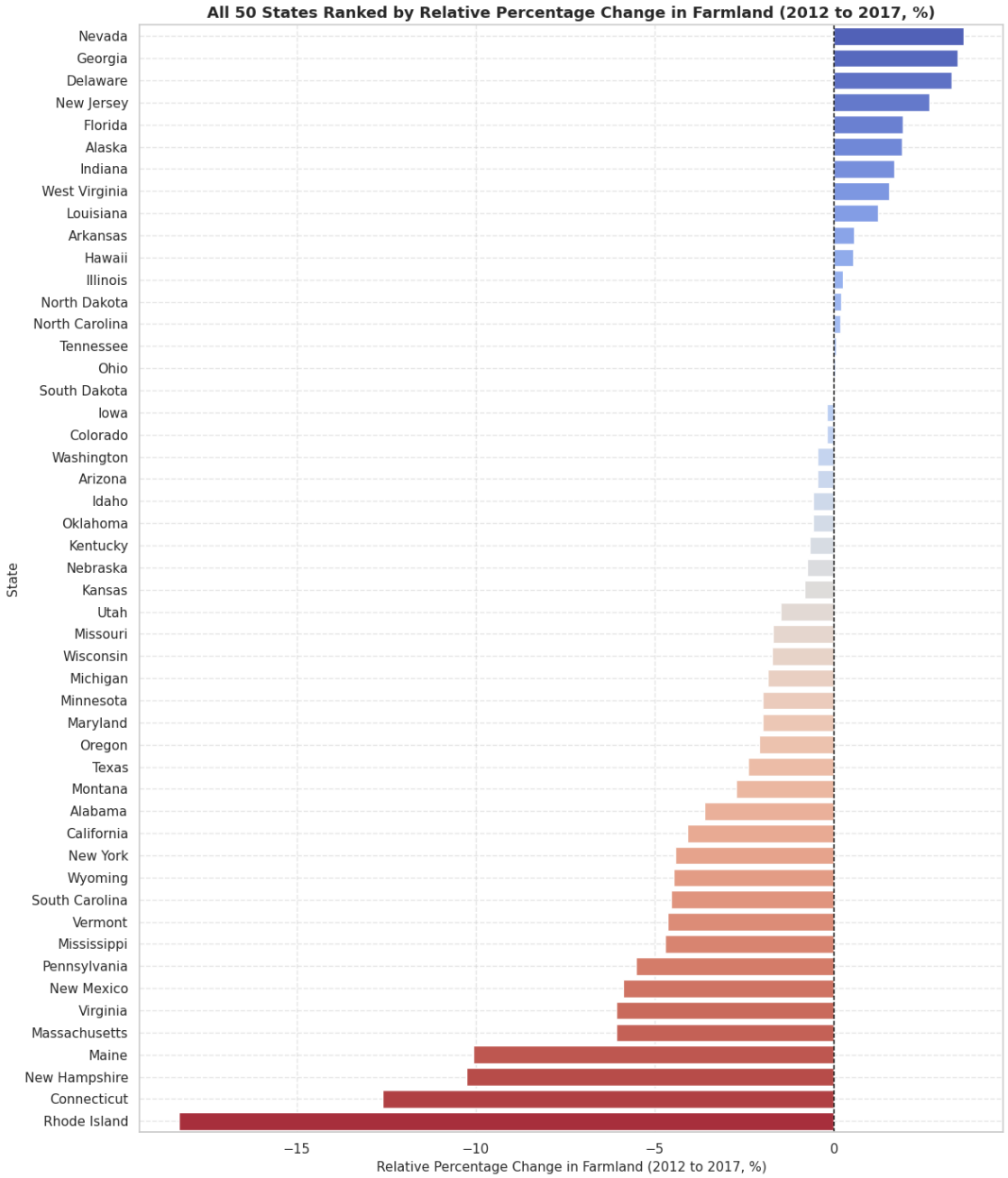
    plt.title(f"All 50 States Ranked by {titles_dict[col]}", fontsize=13,␣
 ↪fontweight='bold')
    plt.xlabel(titles_dict[col], fontsize=11)
    plt.ylabel('State', fontsize=11)
    plt.grid(True, which="major", linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```


All 50 States Ranked by Total Farmland Area in 2012 (Acres)
Texas
Montana
Kansas
Nebraska
South Dakota
New Mexico
North Dakota
Oklahoma
Colorado
Iowa
Wyoming
Missouri
Illinois
Arizona
Minnesota
California
Oregon
Washington
Indiana
Wisconsin
Ohio
Arkansas
Kentucky
Idaho
Utah
Mississippi
Tennessee
Michigan
Georgia
Florida
Alabama
North Carolina
Virginia
Louisiana
Pennsylvania
New York
Nevada
South Carolina
West Virginia
Maryland
Maine
Vermont
Hawaii
Alaska
New Jersey
Massachusetts
Delaware
New Hampshire
Connecticut
Rhode Island
State
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1e8
Total Farmland Area in 2012 (Acres)


All 50 States Ranked by Total Farmland Area in 2017 (Acres)
Texas
Montana
Kansas
Nebraska
South Dakota
New Mexico
North Dakota
Oklahoma
Colorado
Iowa
Wyoming
Missouri
Illinois
Arizona
Minnesota
California
Oregon
Indiana
Washington
Wisconsin
Ohio
Arkansas
Kentucky
Idaho
Tennessee
Utah
Mississippi
Georgia
Michigan
Florida
Alabama
North Carolina
Louisiana
Virginia
Pennsylvania
New York
Nevada
South Carolina
West Virginia
Maryland
Maine
Vermont
Hawaii
Alaska
New Jersey
Delaware
Massachusetts
New Hampshire
Connecticut
Rhode Island
State
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1e8
Total Farmland Area in 2017 (Acres)


All 50 States Ranked by Absolute Change in Farmland (2012 to 2017, Acres)
Georgia
Indiana
Nevada
Florida
Louisiana
North Dakota
Arkansas
Illinois
West Virginia
New Jersey
Delaware
Alaska
North Carolina
Tennessee
Hawaii
Ohio
Rhode Island
South Dakota
Massachusetts
Maryland
New Hampshire
Connecticut
Vermont
Iowa
Colorado
Idaho
Washington
Kentucky
Arizona
Maine
Utah
Michigan
Oklahoma
South Carolina
Wisconsin
New York
Alabama
Oregon
Nebraska
Kansas
Pennsylvania
Missouri
Virginia
Mississippi
Minnesota
California
Wyoming
Montana
New Mexico
Texas
State
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1e6
Absolute Change in Farmland (2012 to 2017, Acres)



2 PART 2: Merging State Landmass Data & Evaluating Agricultural Density

[7]:
```
# 1. Load U.S. Census Gazetteer File containing official land area data
df_state_area = pd.read_csv("2026_Gaz_state_national.csv")
```

```
# Clean column headers
df_state_area.columns = df_state_area.columns.str.strip().str.upper()

# Convert Land Area from Square Miles to Acres (1 Square Mile = 640 Acres)
df_state_area['ALAND_ACRES'] = df_state_area['ALAND_SQMI'] * 640

# Select and rename relevant columns for merging
df_state_area = df_state_area[['NAME','ALAND_ACRES']]
df_state_area.columns = df_state_area.columns.str.lower()

missing_in_gazetteer = set(df_farmland.state) - set(df_state_area.name)
if missing_in_gazetteer:
    print(f"Attention! The following states were not found in the Gazetteer:␣
 ↪{missing_in_gazetteer}")

# 2. Merge Farmland dataset with State Landmass dataset
merged_df = pd.merge(df_farmland,
                     df_state_area,
                     left_on='state',
                     right_on='name',
                     how='left'
                    ).drop('name', axis=1)

# 3. Calculate metrics relative to state and national landmass
# Farmland as % of total state landmass
merged_df['pct_landmass_2012'] = ((merged_df.farmland_2012 / merged_df.
 ↪aland_acres) * 100)
merged_df['pct_landmass_2017'] = ((merged_df.farmland_2017 / merged_df.
 ↪aland_acres) * 100)

# State landmass as % of total U.S. 50-state landmass
total_us_landmass = merged_df.aland_acres.sum()
merged_df['pct_us_aland'] = ((merged_df.aland_acres / total_us_landmass) * 100)

# 4. Display results sorted by highest proportion of state land dedicated to␣
 ↪farming (2017)
print("=> Merged Dataset Ranked by Farmland Share of State Area (2017):\n")
merged_df_sorted = merged_df.sort_values(by='pct_landmass_2017',␣
 ↪ascending=False)

display(merged_df_sorted.assign(
    farmland_2012=lambda df: df['farmland_2012'].map('{:,.0f}'.format),
    farmland_2017=lambda df: df['farmland_2017'].map('{:,.0f}'.format),
    aland_acres=lambda df: df['aland_acres'].map('{:,.0f}'.format),
    diff_abs=lambda df: df['diff_abs'].map('{:+,.0f}'.format),
    diff_pct=lambda df: df['diff_pct'].map('{:+.2f}%'.format),
    pct_landmass_2012=lambda df: df['pct_landmass_2012'].map('{:+.2f}%'.format),
```

```
    pct_landmass_2017=lambda df: df['pct_landmass_2017'].map('{:+.2f}%'.format),
    pct_us_aland=lambda df: df['pct_us_aland'].map('{:+.2f}%'.format)
))
```

=> Merged Dataset Ranked by Farmland Share of State Area (2017):

	state	farmland_2012	farmland_2017	diff_abs	diff_pct	\
26	Nebraska	45,331,783	44,986,821	-344,962	-0.76%	
40	South Dakota	43,257,079	43,243,742	-13,337	-0.03%	
33	North Dakota	39,262,613	39,341,591	+78,978	+0.20%	
15	Kansas	46,137,295	45,759,319	-377,976	-0.82%	
14	Iowa	30,622,731	30,563,878	-58,853	-0.19%	
35	Oklahoma	34,356,110	34,156,290	-199,820	-0.58%	
12	Illinois	26,937,721	27,006,288	+68,567	+0.25%	
42	Texas	130,153,438	127,036,184	-3,117,254	-2.40%	
13	Indiana	14,720,396	14,969,996	+249,600	+1.70%	
24	Missouri	28,266,137	27,781,883	-484,254	-1.71%	
25	Montana	59,758,917	58,122,878	-1,636,039	-2.74%	
34	Ohio	13,960,604	13,965,295	+4,691	+0.03%	
30	New Mexico	43,201,023	40,659,836	-2,541,187	-5.88%	
16	Kentucky	13,049,347	12,961,784	-87,563	-0.67%	
22	Minnesota	26,035,838	25,516,982	-518,856	-1.99%	
5	Colorado	31,886,676	31,820,957	-65,719	-0.21%	
49	Wyoming	30,363,641	29,004,884	-1,358,757	-4.47%	
7	Delaware	508,652	525,324	+16,672	+3.28%	
3	Arkansas	13,810,786	13,888,929	+78,143	+0.57%	
48	Wisconsin	14,568,926	14,318,630	-250,296	-1.72%	
41	Tennessee	10,867,812	10,874,238	+6,426	+0.06%	
2	Arizona	26,249,195	26,125,819	-123,376	-0.47%	
23	Mississippi	10,931,080	10,415,136	-515,944	-4.72%	
46	Washington	14,748,107	14,679,857	-68,250	-0.46%	
19	Maryland	2,030,745	1,990,122	-40,623	-2.00%	
45	Virginia	8,302,444	7,797,979	-504,465	-6.08%	
17	Louisiana	7,900,864	7,997,511	+96,647	+1.22%	
8	Florida	9,548,342	9,731,731	+183,389	+1.92%	
10	Hawaii	1,129,317	1,135,352	+6,035	+0.53%	
32	North Carolina	8,414,756	8,430,522	+15,766	+0.19%	
21	Michigan	9,948,564	9,764,090	-184,474	-1.85%	
9	Georgia	9,620,836	9,953,730	+332,894	+3.46%	
0	Alabama	8,902,654	8,580,940	-321,714	-3.61%	
36	Oregon	16,301,578	15,962,322	-339,256	-2.08%	
37	Pennsylvania	7,704,444	7,278,668	-425,776	-5.53%	
39	South Carolina	4,971,244	4,744,913	-226,331	-4.55%	
4	California	25,569,001	24,522,801	-1,046,200	-4.09%	
47	West Virginia	3,606,674	3,662,178	+55,504	+1.54%	
31	New York	7,183,576	6,866,171	-317,405	-4.42%	
11	Idaho	11,760,109	11,691,912	-68,197	-0.58%	

43	Utah	10,974,396	10,811,604	-162,792	-1.48%
44	Vermont	1,251,713	1,193,437	-58,276	-4.66%
29	New Jersey	715,057	734,084	+19,027	+2.66%
6	Connecticut	436,539	381,539	-55,000	-12.60%
20	Massachusetts	523,517	491,653	-31,864	-6.09%
27	Nevada	5,913,761	6,128,153	+214,392	+3.63%
38	Rhode Island	69,589	56,864	-12,725	-18.29%
28	New Hampshire	474,065	425,393	-48,672	-10.27%
18	Maine	1,454,104	1,307,613	-146,491	-10.07%
1	Alaska	833,861	849,753	+15,892	+1.91%

	aland_acres	pct_landmass_2012	pct_landmass_2017	pct_us_aland
26	49,161,798	+92.21%	+91.51%	+2.17%
40	48,517,944	+89.16%	+89.13%	+2.15%
33	44,156,348	+88.92%	+89.10%	+1.95%
15	52,325,627	+88.17%	+87.45%	+2.31%
14	35,746,383	+85.67%	+85.50%	+1.58%
35	43,901,827	+78.26%	+77.80%	+1.94%
12	35,528,663	+75.82%	+76.01%	+1.57%
42	167,206,078	+77.84%	+75.98%	+7.39%
13	22,928,117	+64.20%	+65.29%	+1.01%
24	43,997,781	+64.24%	+63.14%	+1.95%
25	93,152,190	+64.15%	+62.40%	+4.12%
34	26,149,818	+53.39%	+53.40%	+1.16%
30	77,640,415	+55.64%	+52.37%	+3.43%
16	25,270,932	+51.64%	+51.29%	+1.12%
22	50,964,367	+51.09%	+50.07%	+2.25%
5	66,327,865	+48.07%	+47.98%	+2.93%
49	62,136,673	+48.87%	+46.68%	+2.75%
7	1,247,093	+40.79%	+42.12%	+0.06%
3	33,274,917	+41.51%	+41.74%	+1.47%
48	34,667,693	+42.02%	+41.30%	+1.53%
41	26,383,908	+41.19%	+41.22%	+1.17%
2	72,739,542	+36.09%	+35.92%	+3.22%
23	30,031,899	+36.40%	+34.68%	+1.33%
46	42,532,206	+34.68%	+34.51%	+1.88%
19	6,215,045	+32.67%	+32.02%	+0.27%
45	25,268,074	+32.86%	+30.86%	+1.12%
17	27,661,741	+28.56%	+28.91%	+1.22%
8	34,341,229	+27.80%	+28.34%	+1.52%
10	4,110,459	+27.47%	+27.62%	+0.18%
32	31,119,648	+27.04%	+27.09%	+1.38%
21	36,232,063	+27.46%	+26.95%	+1.60%
9	36,939,283	+26.04%	+26.95%	+1.63%
0	32,417,028	+27.46%	+26.47%	+1.43%
36	61,438,002	+26.53%	+25.98%	+2.72%
37	28,635,125	+26.91%	+25.42%	+1.27%
39	19,241,105	+25.84%	+24.66%	+0.85%

4	99,750,648	+25.63%	+24.58%	+4.41%
47	15,386,470	+23.44%	+23.80%	+0.68%
31	30,159,023	+23.82%	+22.77%	+1.33%
11	52,893,114	+22.23%	+22.10%	+2.34%
43	52,861,210	+20.76%	+20.45%	+2.34%
44	5,899,052	+21.22%	+20.23%	+0.26%
29	4,707,222	+15.19%	+15.59%	+0.21%
6	3,099,229	+14.09%	+12.31%	+0.14%
20	4,992,639	+10.49%	+9.85%	+0.22%
27	70,310,647	+8.41%	+8.72%	+3.11%
38	661,692	+10.52%	+8.59%	+0.03%
28	5,730,556	+8.27%	+7.42%	+0.25%
18	19,740,889	+7.37%	+6.62%	+0.87%
1	365,701,848	+0.23%	+0.23%	+16.17%

3 Part 3: Data Analysis & Policy Synthesis Report

3.1 Data Accuracy, Research, and Methodological Considerations

Careful data checking is important in data analysis to make sure the results are accurate, reliable, and fair. For this study, the main data comes from the USDA Census of Agriculture and the American Farmland Trust (AFT). Several important factors must be considered:

- **Data Sources and Accuracy:** The main farmland data comes from the USDA National Agricultural Statistics Service (NASS). The USDA conducts a Census of Agriculture every five years, in years ending in 2 and 7. Because the census follows strict federal guidelines, its data is a trusted source for information about farmland in the United States. Additionally, total state land area data was gathered from the official U.S. Census Bureau Gazetteer Files, providing authoritative geographic measurements necessary to calculate each state's agricultural land percentage relative to its total landmass.
- **Data Cleaning and Preparation:** Raw farmland data required extensive cleaning: stripping comma thousand-separators, casting string objects to integers, normalizing state names via title casing (resolving entries like `OHIO` to `Ohio`), and calculating accurate absolute and percentage changes before any rounding. Merging with state landmass required strict validation (`how='left'` with post-merge verification) to ensure no states were dropped or duplicated.
- **Data Use, Licensing, and Ethics:** USDA Census data is generally available to the public and can be used freely. However, reports and summaries created by organizations such as the American Farmland Trust may have their own rules about how the data can be used, including restrictions on commercial use. Using public data to study important issues, such as the loss of farmland, supports research and helps the public better understand these challenges. It is important to use the data responsibly and follow any applicable rules.

3.2 Key Findings: Status of Farmland in the USA

- **National Decline:** Overall U.S. farmland decreased by **14,310,081 acres (-1.56%)** between 2012 and 2017, demonstrating persistent land conversion and consolidation across the agricultural sector.

- **Geographic Variations:** Agricultural land density varies widely across the country. States in the Great Plains dedicate the largest proportions of their landmass to agriculture, led by `Nebraska (91.51%)`, `South Dakota (89.13%)`, and `North Dakota (89.10%)`. Conversely, the most severe relative percentage declines occurred in northeastern states, notably `Rhode Island (-18.29%)`, `Connecticut (-12.60%)`, and `New Hampshire (-10.27%)`, largely driven by suburban encroachment.
- **The Ohio Context:** In 2017, Ohio had **13,965,295 acres** of farmland, representing **53.40%** of its total landmass (ranking 19th nationally in agricultural land proportion). Between 2012 and 2017, Ohio experienced a slight net gain of **4,691 acres (+0.03%)**. However, rapid urbanization within Ohio's seven major metropolitan regions, most notably the Columbus metropolitan area, exerts intense development pressure on prime, versatile cropland at city peripheries.

3.3 Key Policy Questions

1. **Protecting Farmland:** How well do state programs that pay landowners to protect farmland (PACE programs) and agricultural zoning laws prevent farmland from being turned into housing, businesses, and other developments, especially near growing cities?
2. **Fairness and Support for New Farmers:** As farmers get older and retire, what financial support, loans, or other incentives could help young and beginning farmers buy or rent farmland instead of having it sold to developers?
3. **Balancing Urban Growth and Farmland Protection:** How can city planners make room for new homes and expanding cities while protecting important farmland? This may include building homes in existing urban areas, encouraging higher-density development, and directing new construction away from the most valuable agricultural land.

3.4 Data Source / Citation:

U.S. Farmland Data (2012 & 2017):

U.S. Department of Agriculture (USDA), National Agricultural Statistics Service (NASS). *2017 Census of Agriculture: Land in Farms by State* (Compiled by American Farmland Trust / Farmland Information Center). (`Farmland.csv`).
URL: https://farmlandinfo.org/statistics/census-of-agriculture/

U.S. State Land Area Data:

U.S. Census Bureau. Geography Division. *2026 State Gazetteer Files: National State Gazetteer File* (`2026_Gaz_state_national.csv`).
URL: https://www.census.gov/geographies/reference-files/time-series/geo/gazetteer-files.2026.html#list-tab-264479560.

[ ]: